Software Project Management Plan (SPMP)

Project Name: Online Food Ordering System for QAU Huts

Prepared According to: IEEE Std 1058-1998

Institution: Quaid-i-Azam University, Islamabad

Date: September 2026

Project Team:

- Muhammad Faheem
- M Saad
- Shahzad Hussain Bakhtawar
- Faizan Ahmad

1. Project Overview

Project Summary

The Online Food Ordering System for QAU Huts is a web and mobile-accessible platform designed to digitize dining services across the Quaid-i-Azam University campus. It reduces physical queues by offering digital menu browsing, remote ordering, campus delivery or pickup selection, status tracking, and digital payments.

Purpose, Scope and Objectives

Purpose: Digitize campus food ordering, streamline hut management, and provide reliable pickup/delivery logistics within QAU.

Scope:

- In-Scope: Account authentication, item catalog browsing, digital cart management, localized campus delivery, online/cash payments, tracking, order cancellations, and admin interfaces for hut management.
- Out-of-Scope: Deliveries outside university bounds, core university payroll/finance, and offline access.

Objectives:

- Minimize ordering wait times across campus huts.
- Enable transparent menu pricing, category browsing, and order-state alerts.
- Deploy a functional, secure system within the designated academic schedule.

Assumptions and Constraints

Assumptions: Users have internet-enabled smartphones/browsers; hut vendors maintain active inventory updates and staff for orders.

Constraints: Academic semester delivery timeframe; operations limited strictly to the QAU Islamabad boundary; reliance on local payment APIs and active internet connectivity.

Project Deliverables

1. Software Requirements Specification (SRS)
2. Software Project Management Plan (SPMP)
3. Architecture & UML Design Artifacts
4. Fully Integrated Web/Mobile Application
5. Test Cases and Execution Reports
6. Final Presentation and System Demonstration

Schedule Summary

- Weeks 1–2 (Sprint 0): Scope Definition, Backlog Setup & Initiation
- Weeks 3–4 (Sprint 1): System Architecture & Database Design
- Weeks 5–6 (Sprints 2–4): Core Module Implementation
- Week 7 (Sprint 5): Functional, Unit, and Acceptance Testing
- Week 8 (Sprint 6): Deployment, Final Review, and Academic Submission

2. References

- IEEE Std 1058-1998, IEEE Standard for Software Project Management Plans.
- ISO/IEC/IEEE 16326, Systems and software engineering — Life cycle processes — Project management.
- Online Food Ordering System for QAU Huts, Software Requirements Specification (SRS), Version 2, 2026.
- Pressman, R. S., Software Engineering: A Practitioner's Approach.
- Mohapatra, P., Software Engineering: A Lifecycle Approach.

3. Definitions

SPMP: Software Project Management Plan.

SRS: Software Requirements Specification.

WBS: Work Breakdown Structure; a hierarchical decomposition of total project scope.

Hut: Authorized on-campus food and beverage kiosks across the QAU Islamabad campus.

Sprint: Time-boxed iterative development cycle in the Agile lifecycle.

NFR: Non-Functional Requirement governing performance, reliability, and security.

4. Project Context

Process Model

The project follows an Agile / Incremental Sprint Model. The life cycle comprises an initial requirements baseline (Sprint 0), design baseline (Sprint 1), multiple two-week code implementation sprints (Sprints 2–4), dedicated verification (Sprint 5), and release retrospectives (Sprint 6).

Methods, Tools and Techniques

Design & Modeling: Use Case Analysis, UML Diagrams (Class, Sequence, Activity), Data Flow Diagrams (DFDs).

Project Management & Scheduling: ProjectLibre for WBS creation, Gantt charting, and resource tracking.

Development & Tracking: Version control with Git/GitHub, IDE environments, and centralized shared storage.

Testing: Unit, functional, integration, and user-acceptance test runs.

Product Acceptance Plan

Acceptance requires verified fulfillment of functional criteria and non-functional targets defined in the SRS:

- Successful student/admin account creation, login, and profile security.
- Complete ordering pipeline: cart selection, transaction approval, delivery routing, and post-order ratings.
- Page loads within 3 seconds under campus network conditions (NFR-01).
- Verification of 57 test cases spanning unit, functional, and acceptance suites.

5. Project Planning

Project Work Plans

The project is partitioned into distinct phases ensuring end-to-end traceability from planning to submission:

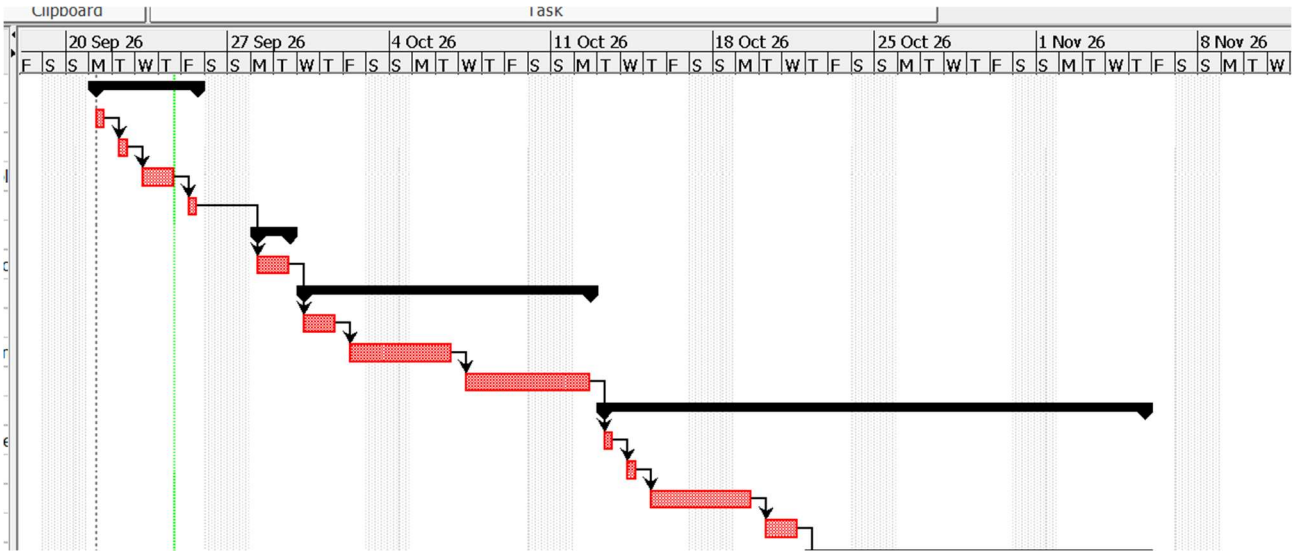
- 1.0 Planning & Backlog: Project charter, stakeholder definition, setup.
- 2.0 Requirements Engineering: Functional matrix, SRS finalization.
- 3.0 System Architecture: DB schema, UI wireframes, UML modeling.
- 4.0 Implementation: Auth, Catalog, Cart/Orders, Delivery, Feedback modules.
- 5.0 Quality Verification: Execution of test cases and bug remediation.
- 6.0 Delivery & Packaging: User manuals, code freeze, final evaluation.

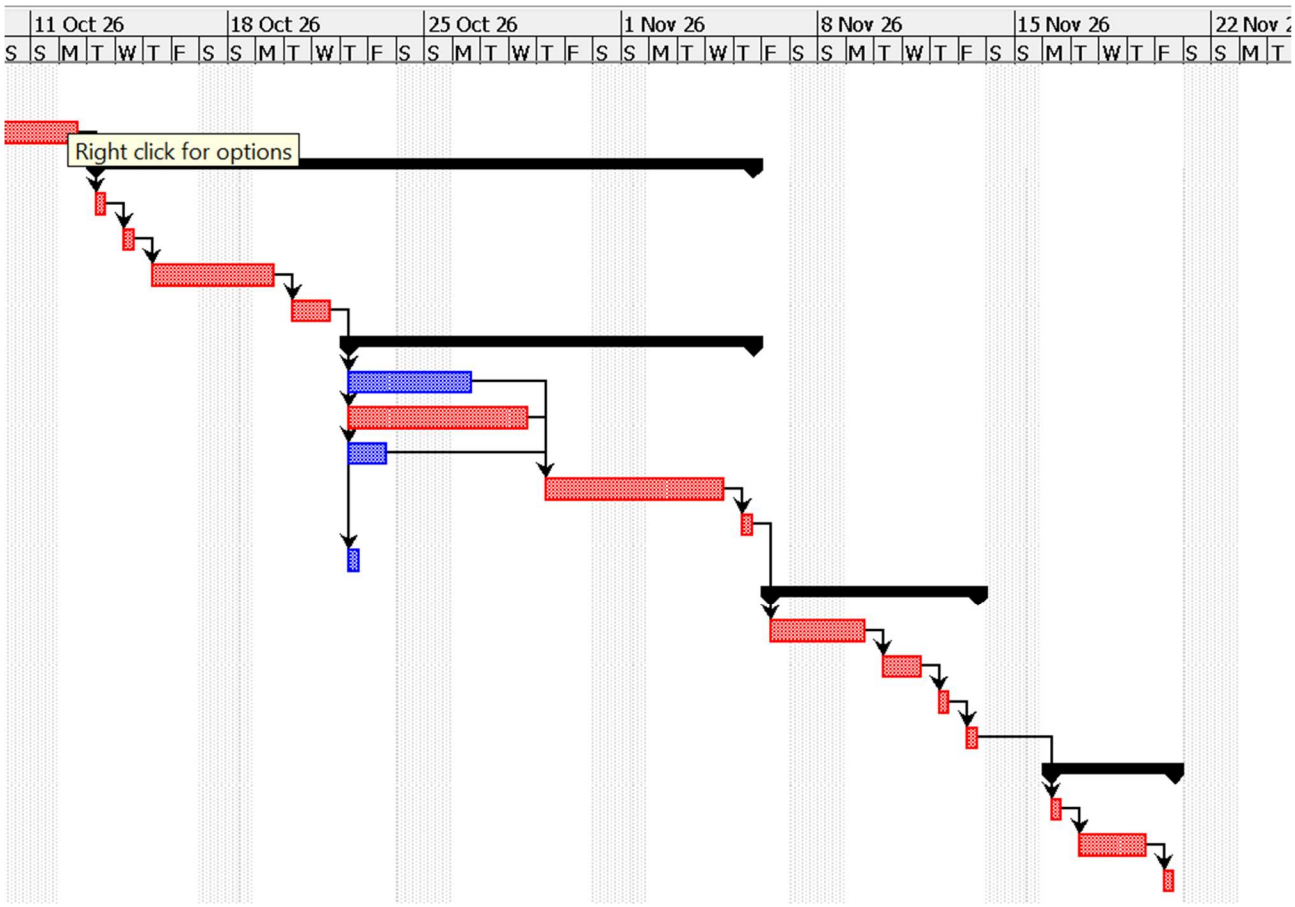
Work Activities

- Activity 1: Requirements elicitation, scope bounding, and SRS drafting.
- Activity 2: UI/UX wireframing, component diagrams, and relational schema creation.
- Activity 3: Backend REST API design, database migrations, and authentication flows.
- Activity 4: Frontend client development for web and mobile interfaces.
- Activity 5: Integration of local payment workflows and delivery notifications.
- Activity 6: Acceptance testing, documentation audit, and final project demonstration.

Schedule Allocation

	Task Name	Duration	Start	Finish
	Planning	5 days	**9/21/26, 8:00 AM**	**9/25/26, 5:00 PM**
	Feasibility study task	1 day	9/21/26, 8:00 AM	9/21/26, 5:00 PM
	Stakeholders Needs	1 day	9/22/26, 8:00 AM	9/22/26, 5:00 PM
	development of project plan	2 days	9/23/26, 8:00 AM	9/24/26, 5:00 PM
	risk management plan	1 day	9/25/26, 8:00 AM	9/25/26, 5:00 PM
	Requirements	2 days	**9/28/26, 8:00 AM**	**9/29/26, 5:00 PM**
	functional and non functional requirements	2 days	9/28/26, 8:00 AM	9/29/26, 5:00 PM
	Analysis	9 days	**9/30/26, 8:00 AM**	**10/12/26, 5:00 PM**
	requirement analysis	2 days	9/30/26, 8:00 AM	10/1/26, 5:00 PM
	make use case discription	3 days	10/2/26, 8:00 AM	10/6/26, 5:00 PM
	SRS document	4 days	10/7/26, 8:00 AM	10/12/26, 5:00 PM
	Design	18 days	**10/13/26, 8:00 AM**	**11/5/26, 5:00 PM**
	make design plan document	1 day	10/13/26, 8:00 AM	10/13/26, 5:00 PM
	make use case diagrams	1 day	10/14/26, 8:00 AM	10/14/26, 5:00 PM
	make details design	3 days	10/15/26, 8:00 AM	10/19/26, 5:00 PM
	design review	2 days	10/20/26, 8:00 AM	10/21/26, 5:00 PM
	construction	11 days	**10/22/26, 8:00 AM**	**11/5/26, 5:00 PM**
	make functional prototype	3 days	10/22/26, 8:00 AM	10/26/26, 5:00 PM
	make fron end design	5 days	10/22/26, 8:00 AM	10/28/26, 5:00 PM
	Database connectivity	2 days	10/22/26, 8:00 AM	10/23/26, 5:00 PM
	development of classes	5 days	10/29/26, 8:00 AM	11/4/26, 5:00 PM
	code review	1 day	11/5/26, 8:00 AM	11/5/26, 5:00 PM
	make tracibility matrix	1 day	10/22/26, 8:00 AM	10/22/26, 5:00 PM
	Testing	6 days	**11/6/26, 8:00 AM**	**11/13/26, 5:00 PM**
	make tracibility matrix	1 day	10/22/26, 8:00 AM	10/22/26, 5:00 PM
	Testing	6 days	**11/6/26, 8:00 AM**	**11/13/26, 5:00 PM**
	unit testing	2 days	11/6/26, 8:00 AM	11/9/26, 5:00 PM
	integration testing	2 days	11/10/26, 8:00 AM	11/11/26, 5:00 PM
	system testing	1 day	11/12/26, 8:00 AM	11/12/26, 5:00 PM
	acceptance testing	1 day	11/13/26, 8:00 AM	11/13/26, 5:00 PM
	Deployment	5 days	**11/16/26, 8:00 AM**	**11/20/26, 5:00 PM**
	final review of system	1 day	11/16/26, 8:00 AM	11/16/26, 5:00 PM
	Documentation	3 days	11/17/26, 8:00 AM	11/19/26, 5:00 PM
	submission	1 day	11/20/26, 8:00 AM	11/20/26, 5:00 PM





Resource Allocation

Team Allocations:

- **Muhammad Faheem:** System Architecture, Requirements, and Backend Integration.
- **M Saad:** Core Implementation, Version Control, and Testing Suites.
- **Shahzad Hussain Bakhtawar:** UI/UX Design, Frontend Development, and Verification.
- **Faizan Ahmad:** Database Design, API Integration, and Technical Documentation.

		Name	RBS	Type	Initials	Max. Units	Cost Per Use	Base Calendar
1		Muhammad Saad		Work	M	100%	$0.00	Standard
2		Muhammad Faheem		Work	M	100%	$0.00	Standard
3		Shahzad Hussain		Work	S	100%	$0.00	Standard
4		Muhammad Faizan		Work	M	100%	$0.00	Standard
5		Github		Material	G		$0.00	
6		Project Libre		Material	P		$0.00	
7		UI/UX		Material	U		$0.00	
8		Visual Studio Code		Material	V		$0.00	
9		HTML		Material	H		$0.00	
10		CSS		Material	C		$0.00	
11		JavaScript		Material	J		$0.00	
12		SQL		Material	S		$0.00	

8. Supporting Process Plans

Configuration Management: All source code, schema migrations, and documentation are version-controlled using Git repository standards.

Quality Assurance Plan: Continuous peer reviews of documents and code, verifying implementation trace directly to SRS criteria.

Documentation & Verification: Technical artifacts and ProjectLibre schedules are reviewed and updated iteratively after each sprint review.